

MEDRIO EDC

Lab Management Module

End-to-End Test Case Document

Prepared by: Jeevan Paapagaari, QA Architecture Team
Classification: Internal QA Use Only
Date: March 10, 2026

Test Execution Summary

#	Test Area	Module Code	TC Count	Priority
1	Lab Permissions & Role Configuration	LPR	12	High
2	Lab Forms — Creation & Restrictions	LFC	10	High
3	Lab Variables — Types & Settings	LVS	10	High
4	Reference Date Variables	RDV	7	High
5	Lab Creation & Management	LCM	10	High
6	Reference Ranges — Setup & Configuration	RRC	12	Critical
7	Range Determinant Logic & Conditions	RDL	9	Critical
8	Effective Dates — Individual & Bulk	EFD	10	High
9	LOINC Codes & Units	LCU	6	Medium
10	Copy & Deactivate Reference Ranges	CDR	8	Medium
11	Lab Data Entry — Form Interaction	LDE	10	Critical
12	Out-of-Range Flag Logic	ORL	12	Critical
13	Field Information Popup Behavior	FIP	6	High
14	Lab Data Upload (Primary/Alternate/Bulk)	LDU	9	High
15	Lab Data Export (Export Labs & Data Export)	LDE2	8	High
16	Audit Logs — Lab Details & Reference Range	AUD	10	High
17	Deactivate & Reactivate Labs	DRL	7	High
18	Study Migration: DEV → TEST → LIVE	SMG	8	Critical
19	Unsupported Features & Negative Scenarios	NEG	10	Medium
20	Cross-Browser & Concurrent Session Tests	CBC	4	Low

Section 1 — Lab Permissions & Role Configuration

Validates that Lab permissions are correctly enforced per user role across all permission levels.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LPR-01	Verify 'Labs' permission visibility under EDC Admin	User is logged in as Study Admin; Labs checkbox is checked under EDC Admin → Enable	- Navigate to Configure → Roles & Permissions. - Click any role (e.g., 'Data Entry User'). - Click the PERMISSIONS tab. - Scroll to verify 'Labs' section is visible.	The 'Labs' section appears under Permissions tab with all sub-permissions listed (Add/Edit Labs, Deactivate Labs, Read-Only Labs, Add/Edit Reference Ranges).
LPR-02	Verify 'Labs' hidden when EDC Admin flag is disabled	User has access to EDC Admin settings; Labs checkbox is currently unchecked	- Go to EDC Admin → Enable. - Uncheck the 'Labs' checkbox and save. - Navigate to Configure → Roles & Permissions → any role → PERMISSIONS tab.	The 'Labs' section does NOT appear in the Permissions tab. No lab-related permissions are visible to that role.
LPR-03	Add/Edit Labs permission — full capability verification	A test role 'LabEditorRole' exists; Add/Edit Labs permission is enabled for this role; user is logged in with LabEditorRole	- Log in as user with Add/Edit Labs permission. - Navigate to Manage → Lab Management. - Attempt to create a new lab named 'Alpha Lab'. - Edit an existing lab name from 'Beta Lab' to 'Beta Lab Updated'. - Navigate to Deactivated Labs and reactivate 'Gamma Lab'.	User can: create 'Alpha Lab' successfully; save name edit for 'Beta Lab Updated'; reactivate 'Gamma Lab'. No permission-denied errors occur.
LPR-04	Verify Deactivate Labs permission isolates correctly	A role has ONLY Deactivate Labs permission enabled (no Add/Edit Labs)	- Log in as user with only Deactivate Labs. - Navigate to Manage → Lab Management. - Click the three-dot icon next to an active lab. - Verify 'Deactivate Lab' option appears. - Attempt to create a new lab via the + icon.	User can deactivate existing labs. The 'Create' action and editing of lab details are blocked (grayed out or permission denied). Only deactivation action works.
LPR-05	Read-Only Labs permission — no write actions allowed	A role has only Read-Only Labs permission	- Log in with Read-Only Labs role. - Navigate to Manage → Lab Management. - Attempt to click on a lab and edit its name field. - Attempt to click Manage Ranges on any Lab Test. - Attempt to create a new lab.	All fields appear as read-only. Edit, Save, Create, and Manage Ranges options are disabled or not rendered. User can only view lab details and reference ranges.
LPR-06	Add/Edit Reference Ranges — dependency on other Lab permission	A role has ONLY 'Add/Edit Reference Ranges' enabled with no other Lab permissions	- Navigate to Configure → Roles & Permissions. - Enable only 'Add/Edit Reference Ranges' for a test role. - Save and log in as that user. - Navigate to Manage → Lab Management.	System displays a validation warning: 'Add/Edit Reference Ranges requires at least one other Lab permission'. The reference range editing is not accessible.
LPR-07	Form-level permissions under Add/Edit Reference Ranges	Add/Edit Reference Ranges is enabled for a role; multiple Lab forms exist (Form_A, Form_B, Form_C)	- Click 'Edit Forms' next to Add/Edit Reference Ranges. - Set Form_A to 'Read & Write'. - Set Form_B to 'Read Only'. - Set Form_C to 'No Access Hide Form'. - Save and log in as that role user. - Navigate to each lab form reference range screen.	Form_A: reference ranges are editable. Form_B: reference ranges visible but fields are read-only. Form_C: form is completely hidden from the user's view.
LPR-08	Study Admin must explicitly set Lab permissions for own role	Study Admin role currently has no Lab permissions configured	- Log in as Study Admin. - Navigate to Manage → Lab Management. - Verify Lab Management is inaccessible.	Initially: Lab Management is inaccessible to Study Admin. After enabling permission: Study Admin can access and manage labs.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
			4. Go to Configure → Roles & Permissions → Study Admin role → PERMISSIONS. 5. Enable 'Add/Edit Labs'. 6. Re-navigate to Manage → Lab Management.	
LPR-09	Permission change reflects immediately in active session	User A is logged in with Read-Only Labs; Admin is logged in simultaneously	1. User A opens Manage → Lab Management (read-only view). 2. In a separate browser, Admin upgrades User A's role to Add/Edit Labs. 3. User A refreshes the Lab Management page.	After refresh, User A now sees editable fields and the Create/Save buttons. No re-login required for permission update to apply.
LPR-10	Verify multiple Lab permissions can coexist on one role	Admin has access to Configure → Roles & Permissions	1. Open a test role. 2. Enable all four Lab permissions: Add/Edit Labs, Deactivate Labs, Read-Only Labs, Add/Edit Reference Ranges. 3. Save. 4. Log in as that role user. 5. Test adding a lab, deactivating a lab, and editing a reference range.	All four permissions are active simultaneously. User can add, deactivate, read, and edit reference ranges without conflicts.
LPR-11	Verify that non-Lab users cannot access Lab Management menu	A user role has zero Lab permissions enabled	1. Log in as a user with no Lab permissions. 2. Navigate to Manage menu. 3. Inspect whether 'Lab Management' option exists.	'Lab Management' option does not appear in the Manage menu for users with no Lab permissions.
LPR-12	Verify Lab permissions survive study deployment	DEV study has role 'QA_Role' with Add/Edit Labs permission; Deploy to TEST is available	1. In DEV study, confirm 'QA_Role' has Add/Edit Labs permission. 2. Deploy study configuration to TEST. 3. In TEST study, navigate to Configure → Roles & Permissions → QA_Role → PERMISSIONS.	After deployment, QA_Role in TEST study retains the Add/Edit Labs permission. Permissions are preserved across environment deployments.

Section 2 — Lab Forms: Creation & Restrictions

Validates Lab form creation, form-use type assignment, visit selection, and unsupported form restrictions.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LFC-01	Create a new Lab form via Configure → Forms	User has Add/Edit Labs permission; at least one standard form exists in study	1. Navigate to Configure → Forms. 2. Click 'Add / Import / Copy a Form'. 3. Select a form from the '--Select a form--' dropdown (e.g., 'Hematology Panel'). 4. In '--Select a form use--', choose 'Lab'. 5. Click Continue. 6. Select visits: Screening, Week 4, Week 8. 7. Click Add.	Form 'Hematology Panel' is created with form use = 'Lab'. It appears in the forms list with a Lab indicator. The three selected visits are associated with the form.
LFC-02	Verify selecting N/A form use creates standard (non-Lab) form	User is at Configure → Forms → Add / Import / Copy a Form	1. Select a form 'Urinalysis Form'. 2. In '--Select a form use--', choose '--N/A--'. 3. Click Continue and complete form creation.	Form 'Urinalysis Form' is created as a standard form. It does NOT appear in the Lab Management module under LAB TESTS tab for any lab.
LFC-03	Lab form appears in Lab Management LAB TESTS tab for lab selection	Lab form 'CBC Panel' has been created; at least one lab 'City Lab' exists in Lab Management	1. Navigate to Manage → Lab Management. 2. Click 'City Lab' → LAB TESTS tab. 3. Verify 'CBC Panel' appears in the form list with a checkbox.	'CBC Panel' Lab form is listed under City Lab's LAB TESTS tab with a selectable checkbox. Selecting it enables Manage Ranges button.
LFC-04	Verify Randomization form cannot be used as Lab form	A Randomization form exists in the study	1. Navigate to Configure → Forms → Add / Import / Copy a Form. 2. Select the Randomization form. 3. In '--Select a form use--', check if 'Lab' option is available or grayed out.	The 'Lab' option is not available or is grayed out for Randomization form types. System prevents saving a Randomization form as a Lab form.
LFC-05	Verify ePRO form cannot be assigned Lab use type	An ePRO form exists in the study	1. Navigate to Configure → Forms. 2. Attempt to edit an ePRO form.	System blocks or does not offer 'Lab' as a form use option for ePRO

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
			3. In form use settings, try selecting 'Lab' for the ePRO form.	forms. An appropriate restriction message is displayed.
LFC-06	Verify eConsent form is not supported as Lab form	An eConsent form exists in the study	1. Navigate to Configure → Forms. 2. Attempt to set an eConsent form's use type to 'Lab'.	eConsent form cannot be assigned Lab use type. System blocks the action with an appropriate restriction.
LFC-07	Verify Inclusion/Exclusion template form is not supported as Lab form	An Inclusion/Exclusion template form exists in the study	1. Navigate to Configure → Forms. 2. Attempt to set an Inclusion/Exclusion template form to Lab use type.	Inclusion/Exclusion template form cannot be assigned Lab use type. System denies or hides the Lab option for this form type.
LFC-08	Create Lab form via Import Form workflow	User has access to import external form files; a valid form import file exists	1. Navigate to Configure → Forms. 2. Click 'Add / Import / Copy a Form' → choose Import. 3. Upload a valid form file. 4. During import configuration, select 'Lab' as form use type. 5. Complete import.	Imported form is created with Lab form use type. It appears in Lab Management under LAB TESTS tab for lab selection.
LFC-09	Create Lab form via Copy Form workflow	An existing Lab form 'Hematology Panel' exists in the study	1. Navigate to Configure → Forms. 2. Click 'Add / Import / Copy a Form' → choose Copy. 3. Select 'Hematology Panel' as source. 4. Assign the copied form the Lab use type. 5. Complete the copy.	A copy of 'Hematology Panel' is created as a Lab form with its variables intact. Appears in Lab Management and can be selected for range configuration.
LFC-10	Out-of-Range flags display in EDC but NOT in DDC for same Lab form	A Lab form 'Coagulation Panel' has Lab Test Result variable 'PT' with lower=9, upper=12; Data exists with PT=15 entered via DDC	1. Open the Coagulation Panel form in DDC mode for subject S-001. 2. Enter PT = 15 (above upper limit of 12). 3. Save the DDC form. 4. Open the same form for subject S-001 in EDC view.	In DDC: No out-of-range flag icon appears next to PT field. In EDC: The upper Out-of-Range flag icon is displayed next to PT variable, indicating value 15 exceeds the upper limit of 12.

Section 3 — Lab Variables: Types & Settings

Validates variable type compatibility with Lab Test Result and Range Determinant settings, PHI restrictions, and single-setting constraint.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LVS-01	Set Number variable as Lab Test Result on a Lab form	Lab form 'Metabolic Panel' exists; user has Add/Edit Labs permission	1. Navigate to Configure → Forms → 'Metabolic Panel'. 2. Hover over Add → Add User-Defined Variables. 3. Select Type: Number, name: 'Glucose Level'. 4. Click Add → open Variable Editor. 5. Under Lab Data, select 'Lab Test Result'. 6. Click Save and Close.	'Glucose Level' variable is saved with Lab Test Result setting. It appears in Lab Management under 'Metabolic Panel' form's Manage Ranges section for range configuration.
LVS-02	Set Categorical Yes/No variable as Lab Test Result	Lab form 'Imaging Panel' exists	1. Add a Categorical Yes/No variable named 'X-Ray Taken' to 'Imaging Panel'. 2. In Variable Editor, set Lab Data → Lab Test Result. 3. Save and Close. 4. Navigate to Lab Management and select 'Imaging Panel' for a lab. 5. Click Manage Ranges → verify 'X-Ray Taken' is listed.	'X-Ray Taken' is listed in Manage Ranges with Yes/No selection options for setting the expected value, not a numeric range.
LVS-03	Set Categorical Select variable as Range Determinant	Lab form 'CBC Panel' exists with a standard form for demographics	1. Add a Categorical Select variable 'Gender' with options Male/Female/Other to the demographics form. 2. In Variable Editor, set Lab Range → Range Determinant. 3. Save and Close. 4. Navigate to Lab Management → Manage Ranges → reference range editor.	'Gender' variable appears in the Range Determinant dropdown and can be selected. Its values (Male/Female/Other) are selectable when configuring range conditions.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
			5. Click Range Determinant field.	
LVS-04	Set Calculated (Age) variable as Range Determinant	<i>A Calculated variable 'Age' exists in the Medrio EDC Variable Library</i>	1. Add the calculated 'Age' variable from the Variable Library to any form. 2. In Variable Editor, set Lab Range → Range Determinant. 3. Save. 4. Verify in reference range editor that 'Age' is selectable as a Range Determinant.	The calculated 'Age' variable is successfully set as Range Determinant. It appears in the Range Determinant field within the reference range configuration, allowing age-based range stratification.
LVS-05	Verify single Lab setting constraint — cannot set both Lab Test Result AND Range Determinant	<i>Lab form 'Hematology Panel' exists with a Number variable 'WBC Count'</i>	1. Open 'WBC Count' variable in the Variable Editor. 2. Set Lab Data → Lab Test Result. 3. Save. 4. Reopen the Variable Editor. 5. Attempt to additionally set Lab Range → Range Determinant on the same variable.	System prevents setting both options. When Lab Test Result is selected, Range Determinant option is disabled (grayed out) and vice versa. An error message indicates only one setting is allowed per variable.
LVS-06	Verify Lab Test Result variable cannot be marked as PHI	<i>Lab form 'Cardiac Panel' exists with a Number variable 'BNP Level'</i>	1. Open 'BNP Level' variable in Variable Editor. 2. Set Lab Data → Lab Test Result. 3. Attempt to also check the PHI (Protected Health Information) checkbox. 4. Click Save.	System prevents saving. PHI checkbox is either disabled when Lab Test Result is selected, or system throws an error: 'Lab Test Result variables cannot be marked as PHI'.
LVS-07	Verify Range Determinant variable cannot be marked as PHI	<i>A demographics form exists with a variable 'Pregnancy Status' (Categorical Yes/No)</i>	1. In Variable Editor for 'Pregnancy Status', set Lab Range → Range Determinant. 2. Attempt to enable the PHI flag. 3. Click Save.	System blocks saving with error message indicating Range Determinant variables cannot be designated as PHI.
LVS-08	Lab Test Display Label customization and display in Lab Management	<i>Lab form 'Lipid Panel' with Number Lab Test Result variable 'LDL_Measurement' exists</i>	1. Open 'LDL_Measurement' in Variable Editor. 2. In Lab Test Display Label field, enter 'LDL Cholesterol (mg/dL)'. 3. Save and Close. 4. Navigate to Manage → Lab Management → select a lab → LAB TESTS → Manage Ranges.	In Lab Management Manage Ranges view, the variable is displayed as 'LDL Cholesterol (mg/dL)' instead of the system variable name 'LDL_Measurement'.
LVS-09	Numeric variable in grid set as Lab Test Result	<i>Lab form 'Chemistry Panel' contains a pre-filled grid with numeric variables</i>	1. Navigate to Configure → Forms → 'Chemistry Panel'. 2. Open a numeric variable within the grid in Variable Editor. 3. Set Lab Data → Lab Test Result. 4. Save and Close. 5. Navigate to Lab Management and verify it appears in Manage Ranges.	The numeric variable inside the grid is accepted as a Lab Test Result variable and appears in the Manage Ranges view for range configuration.
LVS-10	Range Determinant variable on standard (non-Lab) form is valid	<i>A standard form 'Subject Demographics' exists with a variable 'Age_at_Enrollment'</i>	1. Open 'Age_at_Enrollment' variable in Variable Editor. 2. Set Lab Range → Range Determinant. 3. Save and Close. 4. In Lab Management reference range editor, click Range Determinant field. 5. Verify 'Age_at_Enrollment' is listed despite being on a standard form.	'Age_at_Enrollment' from the standard 'Subject Demographics' form appears in the Range Determinant dropdown and can be selected for use in lab reference ranges.

Section 4 — Reference Date Variables

Validates creation, restrictions, and behavior of Reference Date variables used for effective date comparisons.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
RDV-01	Create a Date variable as Reference Date on a Lab form	<i>Lab form 'Hematology Panel' exists; user has Add/Edit Labs permission</i>	1. Navigate to Configure → Forms → 'Hematology Panel'. 2. Hover over Add → Add User-Defined Variables. 3. Select Type: Date, Name: 'Lab Collection Date'.	'Lab Collection Date' is saved as a Date-type Reference Date variable on the Lab form. It appears in the Lab Management reference range editor's Effective Dates section.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
			4. Click Add → open Variable Editor. 5. Under Lab Date setting, select 'Reference Date'. 6. Ensure Day Required, Month Required, Year Required are all checked. 7. Click Save and Close.	
RDV-02	Create a Date and Time variable as Reference Date	Lab form 'Biochemistry Panel' exists	1. Add a 'Date and Time' variable named 'Sample Timestamp' to the Lab form. 2. Set Lab Date → Reference Date. 3. Ensure full date and time fields are required. 4. Save.	'Sample Timestamp' is accepted as a Reference Date variable of type Date and Time. It can be used for effective date range comparison in Lab Management.
RDV-03	Verify non-Date variable types cannot be set as Reference Date	Lab form 'Metabolic Panel' exists with Number and Text variables	1. Open a Number variable in Variable Editor. 2. Check if 'Reference Date' option appears under Lab Date settings. 3. Repeat for a Text variable.	The 'Reference Date' option under Lab Date is not available for Number and Text variable types. It is only selectable for Date and Date/Time variables.
RDV-04	Verify Reference Date must reside on the same Lab form	Two Lab forms exist: Form_A and Form_B; Reference Date variable exists on Form_B	1. Navigate to Lab Management. 2. Select a lab and go to LAB TESTS tab. 3. Check Form_A. 4. Click Manage Ranges → navigate to Effective Dates section. 5. Verify if the Reference Date variable from Form_B appears.	Only the Reference Date variable residing on Form_A is available for Effective Dates in Form_A's reference ranges. Variables from Form_B are not cross-applicable.
RDV-05	Partial date not allowed for Reference Date variable	Reference Date variable 'Lab Collection Date' exists on a Lab form; Day Required is NOT checked	1. Open 'Lab Collection Date' in Variable Editor. 2. Uncheck 'Day Required', keeping Month and Year Required. 3. Attempt to save.	System prevents saving with error: 'Reference Date variables require full date (Day, Month, and Year must all be required). Partial dates are not supported for Lab Management effective date comparison'.
RDV-06	Reference Date variable cannot be marked as PHI	Lab form with a Reference Date variable 'Sample Date' exists	1. Open 'Sample Date' variable in Variable Editor. 2. Select Reference Date under Lab Date. 3. Attempt to enable the PHI flag. 4. Save.	System blocks the save action. PHI flag is either disabled when Reference Date is selected, or validation error is shown: 'Reference Date variables cannot be marked as PHI'.
RDV-07	Reference Date variable cannot be placed in a grid	Lab form with a grid exists	1. Navigate to Configure → Forms → a Lab form with a grid. 2. Attempt to add a Date variable inside the grid. 3. In Variable Editor, set Lab Date → Reference Date. 4. Attempt to save.	System prevents saving a Reference Date variable inside a grid. Error message indicates: 'Effective date variables cannot be arranged in a grid'.

Section 5 — Lab Creation & Management

Validates creating labs, bulk lab entry, lab details configuration, and navigating Lab Management UI.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LCM-01	Create a single lab via Manage → Lab Management	User has Add/Edit Labs permission; no labs exist in the study yet	1. Navigate to Manage → Lab Management. 2. Type 'Sunrise Diagnostics' in the Create Labs text field. 3. Click Create.	'Sunrise Diagnostics' appears in the left panel under Labs. Labs count shows 'Labs (1)'. DETAILS and LAB TESTS tabs become visible. Three-dot icon and search icon appear.
LCM-02	Create multiple labs simultaneously (bulk lab creation)	User has Add/Edit Labs permission	1. Navigate to Manage → Lab Management. 2. In the Create Labs form, enter on separate lines: - BioReference Laboratories - Quest Diagnostics - LabCorp 3. Click Create.	All three labs are created: 'BioReference Laboratories', 'Quest Diagnostics', 'LabCorp' appear in the left panel. Labs counter updates to reflect total active labs count.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
				Scroll bar appears if more than 3 labs listed.
LCM-03	Add lab details — name, lab code, and address	'Sunrise Diagnostics' lab exists in Lab Management	1. Click 'Sunrise Diagnostics' in the lab list. 2. Click DETAILS tab. 3. Enter Lab Name: 'Sunrise Diagnostics Center' 4. Enter Lab Code: 'SDC-001' 5. Enter Address: '1200 Health Blvd, Austin, TX 78701' 6. Click Save.	Details are saved. Lab name updates in the list. Lab code 'SDC-001' and full address are stored and visible in DETAILS tab on revisit.
LCM-04	Verify Add Lab button (+) creates new labs after initial creation	At least one lab already exists; user has Add/Edit Labs permission	1. Navigate to Manage → Lab Management. 2. Click the + (Add Lab) icon. 3. Enter 'Pacific Clinical Lab' and click Create.	'Pacific Clinical Lab' is added to the lab list alongside existing labs. Labs count increments by 1.
LCM-05	Search for a lab by name using the search icon	Multiple labs exist: 'Sunrise Diagnostics', 'Quest Diagnostics', 'LabCorp'	1. Navigate to Manage → Lab Management. 2. Click the Search icon. 3. Type 'Quest'.	Lab list filters to show only 'Quest Diagnostics'. Other labs are not displayed during the active search. Clearing search restores all labs.
LCM-06	Verify labs cannot be deleted — only deactivated	User has Add/Edit Labs permission; 'LabCorp' lab exists	1. Navigate to Manage → Lab Management. 2. Click the three-dot icon next to 'LabCorp'. 3. Inspect the dropdown menu options.	The only option available is 'Deactivate Lab'. There is no 'Delete Lab' option anywhere in the Lab Management UI.
LCM-07	View Lab Details Audit Log from three-dot menu	At least one lab exists and has had details edited previously	1. Navigate to Manage → Lab Management. 2. Click the three-dot icon next to Save button (upper right). 3. Click 'Lab Details Audit Log'. 4. Verify download initiates.	A Microsoft Excel (.xlsx) file downloads to the device. The file contains audit entries including Lab Name, Object, Action, Old Value, New Value, Date, Time, and Changed By columns.
LCM-08	Export Labs to Excel from three-dot menu	Multiple labs with reference ranges configured exist in the study	1. Navigate to Manage → Lab Management. 2. Click the three-dot icon next to Save. 3. Click 'Export Labs'. 4. Download and open the file.	Excel file downloads containing: a tab for lab details and individual tabs for each lab's reference ranges. Data accurately reflects configured reference ranges.
LCM-09	Verify Labs (#) counter is accurate	5 active labs and 2 deactivated labs exist	1. Navigate to Manage → Lab Management. 2. Check the 'Labs (#)' label at the top of the left panel.	'Labs (5)' is displayed — reflecting only ACTIVE labs. Deactivated labs are NOT counted in the Labs (#) counter.
LCM-10	Select Lab forms under LAB TESTS tab and activate Manage Ranges	'Quest Diagnostics' lab exists; Lab forms 'CBC Panel' and 'Metabolic Panel' exist	1. Click 'Quest Diagnostics' → LAB TESTS tab. 2. Check the box next to 'CBC Panel'. 3. Check the box next to 'Metabolic Panel'. 4. Click 'Manage Ranges'.	Manage Ranges view opens for both selected forms. Lab Test Result variables from each form are listed with the range edit icon. Both forms are listed as separate sections.

Section 6 — Reference Ranges: Setup & Configuration

Validates reference range creation, naming, numeric/categorical range entry, and blank field validations.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
RRC-01	Create reference range with numeric lower and upper limits	'BioReference Lab' has 'CBC Panel' with Lab Test Result variable 'WBC Count' (Number type)	1. Navigate to Manage → Lab Management → BioReference Lab → LAB TESTS. 2. Check CBC Panel → Manage Ranges. 3. Click the edit icon next to 'WBC Count'. 4. Enter Name: 'WBC Normal Adult Range'. 5. Enter Lower Limit: 4.5, Upper Limit: 11.0. 6. Click Save icon.	Reference range 'WBC Normal Adult Range' is saved with Lower Limit = 4.5 and Upper Limit = 11.0. Range name appears under WBC Count in the Manage Ranges view.
RRC-02	Verify range name must be unique per Lab Test Result variable	'WBC Normal Adult Range' already exists for 'WBC Count' in 'BioReference Lab'	1. Click + icon to add another reference range for 'WBC Count'. 2. Enter Name: 'WBC Normal Adult Range' (duplicate). 3. Click Save icon.	System displays validation error: 'Range name must be unique for this variable'. Save is blocked until a different name is entered.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
RRC-03	Verify blank numeric range limits trigger validation error	<i>'Quest Diagnostics' has a Number variable 'Platelet Count' as Lab Test Result</i>	1. Navigate to reference range editor for 'Platelet Count'. 2. Enter Name: 'Platelet Normal Range'. 3. Leave Lower Limit and Upper Limit fields empty. 4. Click Save icon.	System shows 'Invalid Entry. Hover for Details' error on both blank limit fields. Save is prevented. Default of 0–0 is shown when the range is first opened.
RRC-04	Default numeric range values pre-populated as 0	<i>Reference range editor is open for a numeric Lab Test Result variable</i>	1. Click + icon to create a new reference range for a Number variable. 2. Observe the Lower Limit and Upper Limit fields before entering any values.	Both Lower Limit and Upper Limit fields show a default value of 0 (not blank). User must update these to valid range values.
RRC-05	Set reference range for Categorical Yes/No Lab Test Result	<i>'Imaging Panel' Lab form has 'X-Ray Taken' as Categorical Yes/No Lab Test Result</i>	1. Navigate to reference range editor for 'X-Ray Taken'. 2. Enter Name: 'X-Ray Expected'. 3. In range values section, select 'Yes'. 4. Click Save icon.	Range 'X-Ray Expected' is saved with expected value = Yes. During data entry, if 'No' is entered, an out-of-range flag will trigger.
RRC-06	Create multiple reference ranges for the same Lab Test Result variable	<i>'CBC Panel' has 'RBC Count' as a Lab Test Result Number variable</i>	1. Create reference range 'RBC Adult Male' (Lower: 4.7, Upper: 6.1). 2. Click the + icon to add another range. 3. Create 'RBC Adult Female' (Lower: 4.2, Upper: 5.4). 4. Save both.	Both 'RBC Adult Male' and 'RBC Adult Female' reference ranges are saved and listed under 'RBC Count'. Each has independent limit values.
RRC-07	Add multiple reference ranges using stratified age groups	<i>'CBC Panel' with Lab Test Result 'Hemoglobin'; Range Determinant variable 'Age' exists</i>	1. Create range 'Hemoglobin Pediatric 0-18' with determinant Age 0–18. 2. Create range 'Hemoglobin Early Adult 19-35' with determinant Age 19–35. 3. Save both. 4. Enter subject data with Age=10 and Hemoglobin=10 (below adult range but within pediatric range).	Only the 'Hemoglobin Pediatric 0-18' range triggers for the 10-year-old subject. 'Hemoglobin Early Adult 19-35' does not trigger. Out-of-range evaluation is age-stratified correctly.
RRC-08	Verify Lab form must be selected before Manage Ranges is available	<i>'Quest Diagnostics' lab exists; LAB TESTS tab is open but no forms are checked</i>	1. Navigate to Quest Diagnostics → LAB TESTS tab. 2. Click 'Manage Ranges' without checking any Lab form checkbox.	'Manage Ranges' button is disabled or an error appears: 'Please select at least one Lab form'. No range editor opens until a form is selected.
RRC-09	Reference range configuration locked after deployment to child study	<i>DEV study has 'CBC Panel' with Lab Test Result variable deployed to TEST</i>	1. In TEST study, navigate to Configure → Forms → CBC Panel. 2. Open the Lab Test Result variable 'WBC Count'. 3. Attempt to change its lab setting from 'Lab Test Result' to 'None'.	System shows warning/error: 'This variable has been used in Lab Management or deployed to a child study. Configuration settings cannot be changed'. Edit is blocked.
RRC-10	Delete and recreate Lab Test Result variable to reset reference ranges	<i>Lab Test Result variable 'Lymphocytes' is used in reference ranges that need to be removed</i>	1. Navigate to Configure → Forms → CBC Panel. 2. Delete the 'Lymphocytes' variable. 3. Observe reference range status in Lab Management. 4. Create a new 'Lymphocytes_v2' variable with Lab Test Result setting. 5. Navigate to Lab Management and configure new ranges.	After deleting 'Lymphocytes': all its reference ranges are deactivated (not deleted). After creating 'Lymphocytes_v2': user can configure fresh reference ranges starting from default 0–0.
RRC-11	Recommend using Range Determinant for unit-based ranges	<i>'Thyroid Panel' has 'TSH' as Lab Test Result; units variable 'TSH_Units' with mIU/L or uIU/mL exists</i>	1. Set 'TSH_Units' as a Range Determinant variable. 2. Create two reference ranges: one for mIU/L (0.4–4.0) and one for uIU/mL (0.4–4.0 converted). 3. In the reference range setup, set the Range Determinant to the unit value used at a specific lab. 4. Save both ranges.	Two unit-specific reference ranges exist for TSH. During data entry, the system checks which unit the lab uses (via the Range Determinant) and applies the correct range. Out-of-range flags respect unit selection.
RRC-12	Verify export labs file can identify overlapping/conflicting	<i>Multiple reference ranges exist for 'WBC Count' that overlap in values (e.g.,</i>	1. Set up overlapping ranges for WBC Count. 2. Navigate to three-dot menu → Export Labs.	System does NOT display any warning or error about overlapping ranges in the UI. The export file lists

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
	ranges (no system warning)	<i>Range A: 3.0–7.0; Range B: 5.0–10.0)</i>	3. Download and open Excel file. 4. Review reference ranges in the export.	all ranges including the overlapping ones. QA must manually review the export to identify conflicts.

Section 7 — Range Determinant Logic & Conditions

Validates that Range Determinant variables correctly stratify reference ranges and that flags trigger only under matching conditions.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
RDL-01	Range Determinant from non-Lab form is selectable in reference range editor	<i>Standard form 'Demographics' has 'Gender' as Categorical Select Range Determinant; 'CBC Panel' Lab form exists</i>	1. Navigate to Lab Management → select a lab → LAB TESTS → check CBC Panel → Manage Ranges. 2. Click edit icon for 'WBC Count'. 3. Click Range Determinant field. 4. Scroll to locate 'Gender' from 'Demographics' form.	'Gender' from the Demographics form is listed in the Range Determinant dropdown. It can be selected for stratifying reference ranges by gender.
RDL-02	Set Gender Range Determinant values for gender-specific ranges	<i>'WBC Count' reference range editor is open; 'Gender' Range Determinant is selected</i>	1. Select 'Gender' as Range Determinant. 2. Set value to 'Female', visit: Screening. 3. Set lower limit: 4.0, upper: 10.5. 4. Save as 'WBC Female Range'. 5. Create another range with Gender = Male, lower: 4.5, upper: 11.0. 6. Save as 'WBC Male Range'.	Both gender-stratified ranges are saved. During data entry, if Gender = Female is on file, 'WBC Female Range' (4.0–10.5) is applied. If Gender = Male, 'WBC Male Range' (4.5–11.0) applies.
RDL-03	Out-of-Range flag triggers ONLY when Determinant condition matches	<i>Subject S-002 has Gender = Female; WBC Female Range 4.0–10.5 exists; WBC Male Range 4.5–11.0 exists</i>	1. Open the CBC Panel Lab form for subject S-002. 2. Select lab 'Quest Diagnostics'. 3. Enter WBC Count = 11.5. 4. Save the form.	The higher out-of-range flag appears next to WBC Count (11.5 exceeds Female upper limit of 10.5). The Male range does NOT trigger because Gender = Female, not Male.
RDL-04	No Out-of-Range flag when Range Determinant data is missing	<i>Subject S-003 has NO Gender value recorded; 'WBC Female Range' exists with Gender determinant</i>	1. Open CBC Panel for subject S-003. 2. Select lab 'Quest Diagnostics'. 3. Enter WBC Count = 13 (exceeds both ranges). 4. Save.	No out-of-range flag appears. Since Gender is not populated for S-003, the Range Determinant condition is unmet and no flag triggers, regardless of the WBC value entered.
RDL-05	Determinant visit selection narrows data collection point	<i>'Age' Range Determinant is configured to be checked at 'Screening' visit</i>	1. Set reference range with Age Range Determinant at visit = Screening. 2. For subject S-004, enter Age = 45 at Screening. 3. Enter a lab value at Week 4 visit that would trigger the 45-year-old adult range.	System uses the Age = 45 value from the Screening visit (as configured) for range determination. Out-of-range flag triggers correctly based on age entered at Screening, not Week 4.
RDL-06	'At any time' determinant option — single-visit scenario	<i>Range Determinant variable 'Pregnancy Status' uses 'At any time' option; a single reference range exists</i>	1. In reference range editor, select 'Pregnancy Status' as Range Determinant with option 'At any time'. 2. Create reference range for Pregnancy = Yes. 3. Enter subject data at any visit with Pregnancy Status = Yes and lab value within range. 4. Verify flag behavior.	Since 'At any time' is selected and Pregnancy Status = Yes exists anywhere in the record, the pregnancy-specific reference range is applied and out-of-range evaluation proceeds accordingly.
RDL-07	ePRO form Range Determinant data does NOT trigger Lab Management	<i>A Range Determinant variable 'Gender_ePRO' exists on an ePRO form with value = Female for subject S-005</i>	1. Subject S-005 has Gender_ePRO = Female entered via ePRO form. 2. Reference range configured with Gender_ePRO as Range Determinant (Female). 3. Open CBC Panel in EDC for S-005, select lab, enter WBC = 15 (above range). 4. Save.	No out-of-range flag triggers. Range Determinant values entered via ePRO forms do NOT activate Lab Management functions. Only values entered in Medrio EDC trigger the evaluation.
RDL-08	Both Lab Test Result AND Range Determinant	<i>Reference range 'WBC Female Range' configured; Subject S-006</i>	1. Open CBC Panel for S-006. 2. Select lab but leave WBC Count field empty.	Step 3: No flag when WBC Count is empty. Step 5: Higher out-of-range flag appears since both Range

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
	data required for flag to trigger	<i>has Gender = Female at Screening</i>	3. Save form. 4. Reopen and enter WBC Count = 12.0 (above 10.5 Female upper limit). 5. Save.	Determinant (Gender = Female) and Lab Test Result (WBC = 12.0) are populated and condition is met.
RDL-09	Range Determinant variable from any visit is usable	<i>Range Determinant 'BMI_Category' is on 'Baseline Assessment' form (not a Lab form)</i>	1. In reference range editor, select 'BMI_Category' from 'Baseline Assessment' form. 2. Set visit to 'Baseline'. 3. Create ranges for BMI_Category = Obese and BMI_Category = Normal. 4. Populate BMI_Category for a test subject. 5. Enter a triggering lab value.	System correctly uses 'BMI_Category' from the Baseline Assessment form to determine which reference range to apply, demonstrating that Range Determinants can come from any form, any visit.

Section 8 — Effective Dates: Individual & Bulk

Validates entry of start/end dates for individual reference ranges and bulk application across multiple ranges in the same lab.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
EFD-01	Enter effective dates for an individual reference range	<i>Reference range 'WBC Normal Adult Range' exists; a Reference Date variable is on the form</i>	1. Navigate to Manage → Lab Management → BioReference Lab → LAB TESTS → Manage Ranges. 2. Open 'WBC Normal Adult Range'. 3. In Effective Dates section, enter Start Date: 01/01/2024. 4. Enter End Date: 12/31/2024. 5. Click Save icon.	Effective dates are saved. Start: 01/01/2024, End: 12/31/2024 displayed under the reference range. A Calendar icon appears next to the lab test name in the LAB TESTS tab.
EFD-02	Validate date format enforcement (mm/dd/yyyy only)	<i>Reference range 'WBC Normal Adult Range' effective dates editor is open</i>	1. Attempt to enter Start Date as '2024-01-01' (ISO format). 2. Attempt to enter Start Date as '01-01-2024' (hyphen format). 3. Attempt the correct format '01/01/2024'.	Entries in non-mm/dd/yyyy format are rejected or not accepted by the date field. Only '01/01/2024' format is valid and accepted by the system.
EFD-03	Validate start date must precede end date	<i>Effective Dates section for a reference range is open</i>	1. Enter Start Date: 12/31/2024. 2. Enter End Date: 01/01/2024 (end before start). 3. Click Save.	System displays validation error: 'Start date must occur before the end date'. Save is blocked. Fields are highlighted indicating invalid date range.
EFD-04	Validate invalid date format is rejected	<i>Effective Dates section is open for 'RBC Adult Female' range</i>	1. Enter Start Date: 13/01/2024 (month 13 — invalid). 2. Click Save.	System displays 'A valid date must be entered'. Save is blocked. Month 13 is not accepted.
EFD-05	Effective dates are optional — save range without dates	<i>A new reference range 'Platelets Pediatric Range' has been configured with lower/upper limits</i>	1. Configure the reference range completely with range values. 2. Leave Start Date and End Date fields blank in Effective Dates. 3. Click Save icon.	Reference range is saved successfully without effective dates. No validation error is triggered. Calendar icon does NOT appear next to lab test name.
EFD-06	Open Effective Dates bulk panel and apply to all ranges	<i>BioReference Lab has 5 reference ranges across CBC Panel and Metabolic Panel forms</i>	1. Navigate to BioReference Lab → LAB TESTS tab. 2. Click 'Effective Dates' button. 3. Select 'All Ranges' checkbox. 4. Enter Start Date: 01/01/2025, End Date: 12/31/2025. 5. Click Apply.	Effective dates 01/01/2025–12/31/2025 are applied to all 5 reference ranges in BioReference Lab. Calendar icons appear next to all lab test names that received dates.
EFD-07	Bulk apply effective dates to all ranges under a single Lab form	<i>BioReference Lab has CBC Panel and Metabolic Panel forms with ranges</i>	1. Open Effective Dates bulk panel. 2. Under 'CBC Panel' section, click 'Select All'. 3. Leave Metabolic Panel ranges unselected. 4. Enter dates 03/01/2025 – 08/31/2025. 5. Click Apply.	Effective dates are applied only to CBC Panel ranges. Metabolic Panel ranges remain without effective dates. Partial selection via 'Select All' under a specific form works as expected.
EFD-08	Search by reference range name in bulk Effective Dates panel	<i>BioReference Lab has multiple ranges including 'WBC Normal Adult</i>	1. Open Effective Dates bulk panel. 2. In Search field, type 'WBC'. 3. Verify filtered results.	Only 'WBC Normal Adult Range' (and any other WBC-named ranges) appear in the filtered list. 'RBC

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
		<i>Range' and 'RBC Adult Female'</i>		Adult Female' and unrelated ranges are hidden during active search.
EFD-09	Warning appears when overwriting existing effective dates via bulk apply	<i>Reference range 'WBC Normal Adult Range' already has effective dates 01/01/2024–12/31/2024</i>	1. Open Effective Dates bulk panel. 2. Select 'WBC Normal Adult Range'. 3. Enter new dates: 06/01/2025 – 12/31/2025. 4. Click Apply.	System displays warning: 'This reference range already has effective dates. Applying will overwrite the existing dates.' After confirming, dates are updated to 06/01/2025–12/31/2025. Audit log captures this overwrite.
EFD-10	Bulk effective dates cannot be cross-applied between different labs	<i>Lab A: BioReference Lab and Lab B: Quest Diagnostics both exist</i>	1. Navigate to BioReference Lab → LAB TESTS → Effective Dates. 2. Select all BioReference ranges. 3. Verify there is no option to apply these dates to Quest Diagnostics ranges in the same action.	The Effective Dates bulk panel is scoped to the currently selected lab only. There is no UI mechanism to apply BioReference Lab's dates to Quest Diagnostics ranges. The user must navigate to Quest Diagnostics separately.

Section 9 — LOINC Codes & Units

Validates entry and export inclusion of LOINC codes and unit values associated with lab test reference ranges.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LCU-01	Enter LOINC code and unit for a reference range	<i>'WBC Count' Lab Test Result variable has an active reference range in BioReference Lab</i>	1. Navigate to Manage → Lab Management → BioReference Lab. 2. Click the reference range name for 'WBC Count'. 3. In Lab Test Unit field, enter 'K/uL'. 4. In LONIC field, enter '26464-8'. 5. Click Save icon.	LOINC code '26464-8' and unit 'K/uL' are saved successfully. They are displayed within the reference range view.
LCU-02	Verify LOINC code appears in Data Export with Include Labs	<i>LOINC code '26464-8' and unit 'K/uL' saved for WBC Count; subject data exists</i>	1. Navigate to Analysis → Data Export. 2. Select CBC Panel Lab form. 3. Check 'Include Labs'. 4. Click Download. 5. Open Excel file and locate WBC Count columns.	Export file includes 'K/uL' unit and '26464-8' LOINC code in the appropriate columns for WBC Count entries.
LCU-03	LOINC code visible in Lab Details Audit Log	<i>LOINC code was added to a reference range</i>	1. Navigate to three-dot menu → Lab Details Audit Log. 2. Download and open Excel file. 3. Search for LOINC code entries.	Audit log contains a row showing the LOINC code addition: Object = 'LOINC', Action = 'Added', New Value = '26464-8', with timestamp and Changed By username.
LCU-04	LOINC code visible in Submission Casebook export	<i>Lab study configured with LOINC codes; submission casebook export option is available</i>	1. Generate a Submission Casebook export for the study. 2. Open the export. 3. Navigate to lab data section.	Submission Casebook includes the LOINC codes and units for all lab test result variables that have them configured in Lab Management.
LCU-05	Update existing LOINC code and verify audit trail	<i>WBC Count reference range has LOINC '26464-8' already saved</i>	1. Open the reference range for WBC Count. 2. Change LOINC from '26464-8' to '6690-2'. 3. Save. 4. Download Reference Range Audit Log.	Audit log shows: Object = 'LOINC', Action = 'Updated', Old Value = '26464-8', New Value = '6690-2'. Change is correctly recorded with timestamp.
LCU-06	Enter units only (no LOINC code) and verify save	<i>Reference range editor for 'Hemoglobin' is open</i>	1. Enter Lab Test Unit: 'g/dL'. 2. Leave LOINC field blank. 3. Click Save.	Reference range is saved with unit 'g/dL' and no LOINC code. No validation error occurs. The unit appears in the export files.

Section 10 — Copy & Deactivate Reference Ranges

Validates copying a reference range preserves all attributes and that deactivation is permanent.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
CDR-01	Copy a reference range and verify all attributes are preserved	BioReference Lab has reference range 'CD4 Female Low' with: determinant Gender=Female, lower=100, upper=600, effective dates 01/01/2024–12/31/2024, unit cells/uL, LOINC 24467-3	1. Navigate to Manage → Lab Management → BioReference Lab. 2. Click the three-dot icon next to 'CD4 Female Low'. 3. Click 'Copy Reference Range'. 4. Open the copied range and verify all fields.	A new range 'CD4 Female Low (Copy)' or similar is created. All attributes match the source: Gender=Female, lower=100, upper=600, effective dates 01/01/2024–12/31/2024, unit=cells/uL, LOINC=24467-3.
CDR-02	Copied reference range stays within the same lab	BioReference Lab reference range 'RBC Adult Male' is being copied	1. Copy 'RBC Adult Male' in BioReference Lab. 2. Navigate to Quest Diagnostics lab. 3. Check if the copied range appears there.	The copy of 'RBC Adult Male' is ONLY visible within BioReference Lab. Quest Diagnostics does NOT have the copied range. Lab isolation of copies is enforced.
CDR-03	Edit copied reference range without affecting the original	'CD4 Female Low (Copy)' has been created from 'CD4 Female Low'	1. Open the copied range 'CD4 Female Low (Copy)'. 2. Change lower limit from 100 to 150. 3. Save. 4. Open original 'CD4 Female Low' and verify its lower limit.	Original 'CD4 Female Low' retains lower limit = 100. The copy's lower limit is updated to 150 independently. Changes to the copy do not affect the source.
CDR-04	Deactivate a reference range	Reference range 'Hemoglobin Pediatric 0-18' is active in BioReference Lab	1. Navigate to the reference range 'Hemoglobin Pediatric 0-18'. 2. Click the three-dot icon next to the range name. 3. Click 'Deactivate Reference Range'. 4. Confirm deactivation if prompted.	Reference range 'Hemoglobin Pediatric 0-18' is deactivated. It is no longer applied during data entry. Out-of-range flags for this range no longer trigger for new data.
CDR-05	Deactivated reference range cannot be reactivated	'Hemoglobin Pediatric 0-18' has been deactivated	1. Navigate to the deactivated range. 2. Click the three-dot icon next to the range name. 3. Check if 'Reactivate Reference Range' option exists.	No 'Reactivate' option is present. The three-dot menu for a deactivated range only shows informational options (or is empty/disabled). Once deactivated, a reference range cannot be reactivated.
CDR-06	Deactivated reference range is visible in audit log	'Platelet Adult Range' has been deactivated	1. Navigate to three-dot menu → Reference Range Audit Log. 2. Download and open Excel. 3. Filter by 'Platelet Adult Range'.	Audit log shows entry: Range Name = 'Platelet Adult Range', Object = 'Range Status', Action = 'Deactivated', New Value = 'Inactive', with timestamp and Changed By.
CDR-07	Copy adds unique name to avoid range name duplication	Reference range 'WBC Normal Adult Range' exists; attempting to copy	1. Copy 'WBC Normal Adult Range'. 2. Note the name given to the copy. 3. Attempt to manually rename the copy to 'WBC Normal Adult Range' (same as original).	The copy is given a unique name (e.g., 'WBC Normal Adult Range Copy' or with a numeric suffix). Renaming the copy to match the original is blocked with: 'Range name must be unique for this variable'.
CDR-08	Deactivated reference range flags remain for historical data	Subject S-010 had an out-of-range flag triggered by 'CD4 Female Low' before its deactivation	1. Deactivate 'CD4 Female Low'. 2. Open subject S-010's CBC form. 3. Check the out-of-range flag status. 4. Export data via Data Export with Include Labs.	Existing flag on S-010's historical record remains visible. Export file retains the flag state. Deactivation only prevents future triggering; historical flags are preserved.

Section 11 — Lab Data Entry: Form Interaction

Validates lab selection requirement, reference range display after save, lab change correction, and data entry restrictions.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LDE-01	Verify Lab dropdown in Lab form is populated with active labs	Three active labs exist: BioReference, Quest, LabCorp; one deactivated lab 'Old Lab' exists	1. Navigate to a subject's CBC Panel Lab form. 2. Click the Lab field at the top of the form. 3. Inspect the dropdown list.	Lab dropdown shows: BioReference, Quest, LabCorp. 'Old Lab' (deactivated) does NOT appear in the selection list.
LDE-02	Verify data cannot be saved without selecting a lab	A subject's CBC Panel Lab form is open with no lab selected	1. Enter WBC Count = 5.0 in the form. 2. Leave Lab field blank. 3. Click Save.	System displays validation error: 'A lab must be selected to save lab form data'. Save is blocked until a lab is selected from the dropdown.
LDE-03	Lab search narrows options as user types	Multiple labs exist with names containing 'Lab': LabCorp, BioReference Lab, Pacific Lab	1. Click Lab field in a CBC Panel form. 2. Type 'bio' in the search field.	Dropdown narrows to show only 'BioReference Lab'. 'LabCorp' and 'Pacific Lab' are hidden as the search filters results in real time.
LDE-04	Reference ranges display as read-only after lab selection and save	Subject S-007 has CBC Panel form; 'BioReference Lab' has WBC Normal range 4.5–11.0	1. Open CBC Panel for S-007. 2. Select 'BioReference Lab' in Lab field. 3. Enter WBC Count = 6.0. 4. Click Save.	After save: reference range '4.5 – 11.0' appears below the WBC Count field as read-only text. The range is visible to the data entry user but cannot be edited on the form.
LDE-05	Reference range shown is specific to the selected lab	BioReference Lab: WBC range 4.5–11.0; Quest Diagnostics: WBC range 4.0–10.8	1. Open CBC Panel for S-008. 2. Select 'BioReference Lab' → save → note WBC reference range displayed. 3. Change lab to 'Quest Diagnostics' → click 'Change Lab' → save. 4. Note the new reference range.	Step 2: WBC range shows '4.5 – 11.0' (BioReference). Step 4: WBC range updates to '4.0 – 10.8' (Quest). Lab-specific ranges are applied correctly.
LDE-06	Correct wrong lab using Change Lab button	Subject S-009 has CBC Panel with BioReference Lab selected and data saved	1. Open CBC Panel for S-009. 2. In Lab field, select 'Quest Diagnostics'. 3. Click 'Change Lab' button next to Save & Next. 4. Verify lab is updated.	'Change Lab' button appears after initial save. After clicking it with Quest Diagnostics selected, the form updates to use Quest Diagnostics. Reference ranges are refreshed to Quest's ranges. Old lab reference ranges are no longer displayed.
LDE-07	Out-of-range flag appears immediately after saving data outside range	BioReference Lab WBC range: 4.5–11.0; Subject S-010 opens CBC Panel	1. Select BioReference Lab. 2. Enter WBC Count = 13.5 (above upper limit 11.0). 3. Click Save.	Immediately after save, the higher out-of-range flag icon appears next to WBC Count (13.5). Reference range '4.5 – 11.0' is displayed below. Flag is visible without page reload.
LDE-08	Data entry user can view out-of-range flags without additional permissions	A data entry user has standard EDC data entry access (no special Lab permission) and accesses a Lab form	1. Log in as data entry user. 2. Open a Lab form with an out-of-range value. 3. Verify flag visibility.	By default, all users with access to Lab forms can view out-of-range flags. No additional permission grant is required. Flags are visible to the data entry user.
LDE-09	Lab form auto-populates Lab field if only one active lab exists	Only one active lab 'SingleLab' exists in the study; all other labs are deactivated	1. Open any Lab form for a new subject. 2. Inspect the Lab field.	System behavior: either Lab field auto-selects 'SingleLab' OR clearly shows it as the only option when the dropdown is opened. User does not have to scroll through multiple options.
LDE-10	Verify multiple form visits use lab-specific ranges at each visit	Subject S-011 CBC Panel visits: Screening (Quest selected), Week 4 (BioReference selected)	1. Open Screening visit CBC Panel → select Quest → enter WBC = 9.5 → save. 2. Open Week 4 visit CBC Panel → select BioReference → enter WBC = 9.5 → save.	Screening visit: WBC 9.5 shows Quest ranges (4.0–10.8) → no flag (within range). Week 4: WBC 9.5 shows BioReference ranges (4.5–11.0) → no flag (within range). Each visit independently uses its selected lab's ranges.

Section 12 — Out-of-Range Flag Logic

Validates numeric boundary conditions, categorical flag behavior, multi-range conflicts, and the Field Information popup.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
ORL-01	Value below lower limit triggers lower out-of-range flag	WBC Count range: 4.5–11.0 in BioReference Lab; subject S-012 CBC Panel form is open	1. Select BioReference Lab. 2. Enter WBC Count = 3.0 (below lower limit 4.5). 3. Save.	The lower out-of-range flag icon appears next to WBC Count. The flag icon visually differs from the upper flag (downward arrow / low indicator). Reference range 4.5–11.0 displayed below the field.
ORL-02	Value above upper limit triggers higher out-of-range flag	WBC Count range: 4.5–11.0; subject S-013 CBC Panel form is open	1. Select BioReference Lab. 2. Enter WBC Count = 15.0 (above upper limit 11.0). 3. Save.	The higher out-of-range flag icon appears (upward arrow / high indicator). Reference range 4.5–11.0 is displayed. Flag icon is visually distinct from the lower flag.
ORL-03	Value exactly at lower boundary — no flag triggered	WBC Count range: 4.5–11.0	1. Enter WBC Count = 4.5 (equal to lower limit). 2. Save.	No out-of-range flag appears. The boundary value 4.5 is treated as within range (inclusive). Reference range is displayed without any flag icon.
ORL-04	Value exactly at upper boundary — no flag triggered	WBC Count range: 4.5–11.0	1. Enter WBC Count = 11.0 (equal to upper limit). 2. Save.	No out-of-range flag appears. The boundary value 11.0 is treated as within range (inclusive). Flag is only triggered strictly beyond the defined limits.
ORL-05	Value returns to within range — flag becomes Inactive	WBC Count = 15.0 was entered; higher flag is Active; subject S-013 form is open	1. Open CBC Panel for S-013. 2. Update WBC Count from 15.0 to 7.0 (within 4.5–11.0). 3. Save. 4. Click the flag icon area and view Field Information popup.	Higher out-of-range flag icon disappears from the form. Field Information popup now shows Status = 'Inactive'. The timestamp of flag resolution and the original activation time are both visible.
ORL-06	Categorical Yes/No: incorrect answer triggers out-of-range flag	X-Ray Taken range: Expected = Yes; Subject S-014 Imaging Panel form is open	1. Select a lab. 2. Enter X-Ray Taken = No. 3. Save.	Out-of-range flag icon appears next to 'X-Ray Taken'. Field Information popup shows: Expected Value = Yes, Entered = No, Status = Active.
ORL-07	Multiple overlapping ranges trigger multiple flags for one variable	WBC Count has two ranges: Range A (3.0–7.0) and Range B (5.0–10.0); entered value = 2.5	1. Enter WBC Count = 2.5. 2. Save. 3. Count the flag icons next to WBC Count. 4. Click the flag icon to view Field Information popup.	Multiple out-of-range flag icons appear (one per triggered range). Field Information popup lists ALL triggered ranges (both Range A and Range B). However, only the FIRST applicable range in the list is shown below the variable on the form.
ORL-08	Field Information popup shows first applicable range below variable	WBC Count: Range A and Range B both trigger; Range A is first in the list	1. Enter WBC Count = 2.5. 2. Save. 3. Observe the reference range text shown below the WBC Count field on the form.	Only Range A's limits are shown below the WBC Count field on the form (the first applicable range). Range B's limits are NOT shown inline — they are only visible in the Field Information popup.
ORL-09	Out-of-Range flag persists for deactivated lab data	Subject S-015 had WBC = 15 flagged under 'Old Lab'; 'Old Lab' was deactivated	1. Deactivate 'Old Lab'. 2. Open CBC Panel for S-015. 3. View WBC Count field.	The out-of-range flag REMAINS on S-015's form even after lab deactivation. Historical flags are preserved. The deactivated lab status does not remove past flags from subject records.
ORL-10	Out-of-Range flag does not appear when no reference range is configured	A Lab Test Result variable 'Neutrophils' has no reference ranges set up in any lab	1. Open CBC Panel for a subject. 2. Select a lab. 3. Enter Neutrophils = 999. 4. Save.	No out-of-range flag appears for Neutrophils since no reference range has been configured. The system only evaluates ranges that are explicitly set.
ORL-11	Flag status in Field Information popup	WBC Count = 15 triggers an out-of-range flag at a known date/time	1. Enter out-of-range value and save. 2. Note the system time. 3. Click the flag icon. 4.	Field Information popup shows the exact timestamp matching the save time. Timestamp format includes

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
	shows correct timestamp		Check 'Time and date the out-of-range flag triggered' in popup.	date and time (e.g., '01/15/2025 14:32:07'). Timestamp is accurate to within 60 seconds.
ORL-12	Multiple subjects: flags evaluated independently per subject	Subject S-016: WBC=13.0 (out of range); Subject S-017: WBC=7.0 (in range); same lab and range configured	1. Enter WBC = 13.0 for S-016 → save. 2. Enter WBC = 7.0 for S-017 → save. 3. Open each subject's form and compare flag states.	S-016: Out-of-range higher flag appears. S-017: No flag appears. Flag evaluation is independent per subject — one subject's flag does not affect another's.

Section 13 — Field Information Popup Behavior

Validates the content, accuracy, and state transitions shown in the Out-of-Range Field Information popup.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
FIP-01	Popup shows all five required data fields for active flag	Subject S-020 has WBC = 15.0 flagged under 'BioReference Lab', range 'WBC Normal Adult Range' 4.5–11.0	1. Open CBC Panel for S-020. 2. Click the higher out-of-range flag icon next to WBC Count.	Field Information popup displays: 1. Range Name: 'WBC Normal Adult Range' 2. Expected Range: 4.5 – 11.0 3. Lab Name: BioReference Lab 4. Flag Status: Active 5. Timestamp: (date/time flag was triggered)
FIP-02	Popup status changes to Inactive when value is corrected	S-020 had WBC = 15.0 Active flag; now updating to 8.0	1. Update WBC Count from 15.0 to 8.0. 2. Save. 3. Click former flag area → open Field Information popup.	Status in popup changes from 'Active' to 'Inactive'. Flag icon disappears from the form. Popup retains original activation timestamp and adds resolution context.
FIP-03	Popup shows all triggered ranges for multi-range flag scenario	WBC Count has Range A and Range B both triggered for value 2.5	1. Click flag icon next to WBC Count. 2. Count the number of range entries listed in the popup.	Both Range A and Range B appear in the Field Information popup with their respective names, expected ranges, and flag statuses. Popup accurately represents all triggered ranges.
FIP-04	Popup is accessible without additional permissions	A data entry user with standard access opens a Lab form with an out-of-range value	1. Log in as data entry user. 2. Open Lab form with flagged variable. 3. Click the flag icon.	Field Information popup opens successfully. Data entry users can view the popup without needing Lab Management-specific permissions.
FIP-05	Popup for categorical Yes/No out-of-range shows Expected Value correctly	X-Ray Taken: Expected = Yes; subject entered No → Active flag	1. Click out-of-range flag icon next to 'X-Ray Taken'. 2. View Expected Range in popup.	Popup shows Expected Value = 'Yes' (not a numeric range). Entered value = 'No'. Status = Active. Lab name and timestamp are also displayed.
FIP-06	Popup timestamps are in study-configured timezone	Study timezone is configured as US Eastern Time (UTC-5)	1. Enter out-of-range value at a known UTC time. 2. Open Field Information popup. 3. Compare popup timestamp to UTC reference.	Popup timestamp reflects the study's configured timezone (US Eastern Time). Timestamp is not displayed in UTC unless the study timezone is UTC.

Section 14 — Lab Data Upload (Primary / Alternate / Bulk)

Validates upload mechanisms for lab data including bulk upload field mapping, multi-subject upload, and out-of-range flag generation post-upload.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LDU-01	Verify Bulk Upload field mapping includes Lab field	Bulk upload is configured in Configure → Bulk Upload for CBC Panel Lab form	1. Navigate to Configure → Bulk Upload → Add new field mapping. 2. Select 'CBC Panel' as the form. 3. Select a valid upload file. 4. Inspect the Form (Field) Name column in the mapping interface.	'Lab' appears as a mandatory field in the Form (Field) Name column. It must be mapped to a column in the upload file before the mapping can be saved.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LDU-02	Map the Lab Name column in upload file to Lab field during Bulk Upload	Upload CSV has a column 'LabName' with values 'BioReference Lab'	1. In the Bulk Upload mapping screen, match 'LabName' from file to 'Lab' field. 2. Complete other field mappings. 3. Save mapping. 4. Upload the file.	Upload succeeds. Subjects' forms are populated with the correct lab (BioReference Lab). Lab dropdown on each affected form reflects the uploaded lab name.
LDU-03	Bulk upload with unmapped Lab field fails with error	Upload CSV is configured but Lab column is not mapped	1. In Bulk Upload mapping, leave Lab field unmapped. 2. Attempt to save or upload.	System displays error: 'Lab field must be mapped before uploading data to a Lab form'. Upload is blocked until Lab is mapped.
LDU-04	Out-of-range flags appear for uploaded lab data	Upload file contains WBC = 14.0 for subject S-025 in BioReference Lab (range 4.5–11.0)	1. Upload the file via Bulk Upload. 2. After upload completes, navigate to CBC Panel form for S-025. 3. Check for out-of-range flag.	Out-of-range higher flag appears next to WBC Count (14.0). Flag triggers correctly for bulk-uploaded data, same as manually entered data.
LDU-05	Upload multiple subjects simultaneously via Bulk Upload	Upload CSV contains valid CBC Panel data for 20 subjects with BioReference Lab	1. Upload the 20-subject CSV via Bulk Upload. 2. After completion, navigate to CBC Panel for random subjects S-001, S-010, S-020.	All 20 subjects have CBC Panel data populated with the correct lab selected. Upload is processed batch without individual entry. Progress indicator shows completion status.
LDU-06	Primary Format Data Upload — Lab data accepted	Primary Format upload file exists with lab test values for subject S-030	1. Navigate to the subject S-030 page. 2. Initiate Primary Format Data Upload. 3. Upload the file. 4. Open CBC Panel for S-030.	Lab data is populated from the Primary Format upload. Lab field is set to the lab specified in the upload file. Out-of-range evaluation runs post-upload.
LDU-07	Alternate Format Data Upload — Lab data accepted	Alternate Format upload file is available for CBC Panel	1. Navigate to Alternate Format Data Upload. 2. Select CBC Panel. 3. Upload valid alternate format file with lab test values. 4. Verify data in forms.	Data is populated in the correct forms via Alternate Format. Lab field is set correctly. Reference ranges and out-of-range flags are evaluated after upload.
LDU-08	Invalid lab name in upload file causes row-level error	Upload CSV contains 'UnknownLab' as lab name for subject S-035 (lab does not exist in study)	1. Map and upload the file. 2. Review upload results.	Row for S-035 fails with error: 'Lab name UnknownLab not found in Lab Management'. Other valid rows upload successfully. Error log identifies the problematic row.
LDU-09	Bulk Upload with deactivated lab name causes validation failure	Upload CSV references 'Old Lab' which has been deactivated	1. Upload file with 'Old Lab' as the lab name. 2. Review upload results.	Rows referencing 'Old Lab' fail with error indicating the lab is not active. Active lab names must be used in uploads.

Section 15 — Audit Logs: Lab Details & Reference Range

Validates that all lab detail changes and reference range modifications are accurately captured in downloadable Excel audit logs.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
AUD-01	Lab Details Audit Log — verify all required columns exist	Multiple lab detail changes have been made (lab name edit, address update, lab code add)	1. Navigate to Manage → Lab Management. 2. Click three-dot icon next to Save. 3. Click 'Lab Details Audit Log'. 4. Download and open the Excel file. 5. Check column headers.	Excel file contains exactly these columns: Lab Name, Object, Action, Old Value, New Value, Date, Time, Changed By. No columns are missing or misordered.
AUD-02	Lab Details Audit Log — records lab name change accurately	Lab 'Sunrise Diagnostics' was renamed to 'Sunrise Diagnostics Center' by user 'qa_admin'	1. Download Lab Details Audit Log. 2. Filter by Lab Name = 'Sunrise Diagnostics Center'. 3. Locate the name change entry.	Audit row shows: Lab Name = Sunrise Diagnostics Center, Object = Lab Name, Action = Updated, Old Value = 'Sunrise Diagnostics', New Value = 'Sunrise Diagnostics Center', Changed By = qa_admin, with correct timestamp.
AUD-03	Lab Details Audit Log — records lab deactivation	Lab 'LabCorp' was deactivated by user 'study_manager'	1. Download Lab Details Audit Log. 2. Search for deactivation entry.	Audit row shows: Object = 'Lab Status', Action = 'Deactivated', Old Value = 'Active', New Value =

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
				'Inactive', Changed By = study_manager.
AUD-04	Lab Details Audit Log — records lab reactivation	Lab 'LabCorp' was reactivated after deactivation	1. Download Lab Details Audit Log. 2. Find reactivation entry for LabCorp.	Audit row shows: Object = 'Lab Status', Action = 'Reactivated', Old Value = 'Inactive', New Value = 'Active', with correct timestamp and user.
AUD-05	Reference Range Audit Log — verify all required columns	Reference range changes have been made (determinant added, range values edited, dates updated)	1. Navigate to three-dot icon → Reference Range Audit Log. 2. Download and open Excel. 3. Verify column headers.	Excel contains: Lab Name, Lab Form, Lab Test Name, Range Name, Object, Action, Old Value, New Value, Date, Time, Changed By. All columns present.
AUD-06	Reference Range Audit Log — records lower limit change	WBC Normal Adult Range lower limit was changed from 4.5 to 4.0 by 'qa_admin'	1. Download Reference Range Audit Log. 2. Filter by Range Name = 'WBC Normal Adult Range'. 3. Find lower limit change entry.	Audit row: Lab Form = CBC Panel, Lab Test Name = WBC Count, Range Name = WBC Normal Adult Range, Object = Lower Limit, Action = Updated, Old Value = 4.5, New Value = 4.0, Changed By = qa_admin.
AUD-07	Reference Range Audit Log — records effective date addition	Effective dates 01/01/2025–12/31/2025 were added to 'WBC Normal Adult Range'	1. Download Reference Range Audit Log. 2. Find effective date entry for 'WBC Normal Adult Range'.	Audit log shows two rows: one for Start Date (Action = Added, New Value = 01/01/2025) and one for End Date (Action = Added, New Value = 12/31/2025). Both entries reference 'WBC Normal Adult Range'.
AUD-08	Reference Range Audit Log — records determinant addition	Gender Range Determinant was added to 'RBC Adult Female' range	1. Download Reference Range Audit Log. 2. Locate determinant addition entry.	Audit row: Object = 'Range Determinant', Action = 'Added', New Value = 'Gender = Female', with correct lab, form, and range name context.
AUD-09	Both audit logs are downloadable as Excel (.xlsx) files	Lab Management is accessible with valid labs and ranges	1. Download Lab Details Audit Log → verify file extension. 2. Download Reference Range Audit Log → verify file extension.	Both downloads produce .xlsx files that open correctly in Microsoft Excel. Files are not corrupted and all columns are present.
AUD-10	Audit logs captured for LOINC code updates	LOINC code for WBC Count was changed from '26464-8' to '6690-2'	1. Download Lab Details Audit Log. 2. Filter by Object = 'LOINC'.	Audit entry shows: Object = LOINC, Action = Updated, Old Value = 26464-8, New Value = 6690-2. Lab name, Changed By, and timestamp are accurate.

Section 16 — Deactivate & Reactivate Labs

Validates lab deactivation workflow, historical data preservation, and reactivation via Deactivated Labs view.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
DRL-01	Deactivate a lab via three-dot icon	Lab 'Pacific Clinical Lab' is active; user has Deactivate Labs permission	1. Navigate to Manage → Lab Management. 2. Click three-dot icon next to 'Pacific Clinical Lab'. 3. Click 'Deactivate Lab'. 4. Confirm if prompted.	'Pacific Clinical Lab' is removed from the active Labs list. Labs count decrements by 1. Lab no longer appears in data entry Lab field dropdown.
DRL-02	Out-of-range flags remain after lab deactivation	Subject S-030 has WBC = 14.0 out-of-range flag from Pacific Clinical Lab before deactivation	1. Deactivate Pacific Clinical Lab. 2. Open CBC Panel for S-030. 3. Check WBC Count flag status.	Out-of-range flag remains visible on S-030's CBC Panel. Deactivating the lab does NOT clear historical flags. Field Information popup still shows range and lab name.
DRL-03	Deactivated lab appears in Deactivated Labs list	'Pacific Clinical Lab' has been deactivated	1. Navigate to Manage → Lab Management. 2. Click three-dot icon next to Save. 3. Click 'Deactivated Labs'.	Deactivated Labs screen shows 'Pacific Clinical Lab' with its details, selected forms, and reference ranges still visible for reference.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
DRL-04	Reactivate a deactivated lab	<i>'Pacific Clinical Lab' is in the Deactivated Labs list</i>	1. Navigate to Deactivated Labs. 2. Click three-dot icon next to 'Pacific Clinical Lab'. 3. Click 'Reactivate Lab'. 4. Navigate back to active Labs list.	'Pacific Clinical Lab' reappears in the active Labs list. It is available again in the data entry Lab field dropdown. Labs count increments to include the reactivated lab.
DRL-05	Reactivated lab retains all previous details and ranges	<i>'Pacific Clinical Lab' has been reactivated after deactivation</i>	1. Click 'Pacific Clinical Lab' in active Labs. 2. Check DETAILS tab for name, code, address. 3. Check LAB TESTS tab for associated forms and reference ranges.	All previously configured details (lab name, lab code, address) and reference ranges are intact after reactivation. No configuration is lost during the deactivation/reactivation cycle.
DRL-06	Lab cannot be deleted — confirms permanent deactivation as only removal option	<i>Multiple labs exist in Lab Management</i>	1. Search throughout Lab Management UI for any delete/remove button or option for labs. 2. Check right-click menus, three-dot menus, DETAILS tab options.	No delete option exists anywhere in Lab Management. The three-dot icon next to a lab name only shows 'Deactivate Lab'. This is the intended design — labs can only be deactivated, never deleted.
DRL-07	Export in Deactivated Labs shows all historical ranges intact	<i>Pacific Clinical Lab is deactivated; Export Labs is initiated from the global three-dot menu</i>	1. Click three-dot icon next to Save → Export Labs. 2. Open Excel. 3. Look for Pacific Clinical Lab tab.	Export includes a tab for Pacific Clinical Lab with its reference ranges, even though the lab is deactivated. Historical data is exportable for compliance and audit purposes.

Section 17 — Study Migration: DEV → TEST → LIVE

Validates the lab configuration deployment flow across study environments and enforces the correct configuration hierarchy.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
SMG-01	Deploy Lab configuration from DEV to TEST	<i>DEV study has: Lab forms (CBC Panel), Lab variables (WBC Count - Lab Test Result, Gender - Range Determinant), Lab permissions configured for QA_Role</i>	1. In DEV study, verify Lab forms, variables, and permissions are configured. 2. Initiate deploy to TEST study. 3. In TEST study, navigate to Configure → Forms → check CBC Panel. 4. Check Variable Editor for WBC Count and Gender. 5. Check Roles and Permissions for QA_Role.	TEST study has: CBC Panel as Lab form, WBC Count as Lab Test Result, Gender as Range Determinant, QA_Role with same Lab permissions. All configuration is preserved.
SMG-02	Labs and reference ranges are set up in LIVE (not DEV/TEST)	<i>TEST deployment was successful; LIVE study is accessible</i>	1. In DEV study, navigate to Manage → Lab Management → attempt to create a lab. 2. In TEST study, navigate to Manage → Lab Management → attempt to create a lab. 3. In LIVE study, navigate to Manage → Lab Management → create a lab 'Production Lab'.	DEV and TEST: Lab creation is permitted for testing but is the configuration/test environment. LIVE: 'Production Lab' is successfully created. The Labs and reference ranges setup is expected to be done in LIVE for actual usage.
SMG-03	Configuration changes locked for deployed variables	<i>WBC Count (Lab Test Result) in DEV has been deployed to TEST</i>	1. In DEV study, open WBC Count Variable Editor after deployment. 2. Attempt to change its Lab setting from 'Lab Test Result' to None or Range Determinant.	System blocks the change with warning: 'This variable has been deployed to a child study or used in Lab Management. Configuration settings cannot be changed'.
SMG-04	DEV → TEST migration preserves multi-visit Lab form assignments	<i>CBC Panel is assigned to visits: Screening, Week 4, Week 8 in DEV</i>	1. Deploy DEV to TEST. 2. In TEST, navigate to Configure → Forms → CBC Panel. 3. Verify visits assigned to CBC Panel.	TEST study CBC Panel retains all three visits: Screening, Week 4, and Week 8. Visit assignments are preserved through deployment.
SMG-05	Reference ranges configured in TEST do not deploy to LIVE automatically	<i>TEST study has a reference range 'WBC Test Range' set up for internal testing</i>	1. In TEST, create reference range 'WBC Test Range' with values 3.0–13.0 (wider than production). 2. Deploy TEST configuration to LIVE. 3. In LIVE, navigate to Manage → Lab Management → check for 'WBC Test Range'.	Reference ranges (created in Manage → Lab Management) are NOT part of the configuration deployment. LIVE study does not receive 'WBC Test Range'. Reference ranges are managed independently in each environment.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
SMG-06	LIVE study: create production reference ranges independently	<i>LIVE study has 'Production Lab' created; CBC Panel selected in LAB TESTS tab</i>	1. In LIVE, navigate to Lab Management → Production Lab → LAB TESTS → Manage Ranges. 2. Create 'WBC Adult Production Range' with Lower: 4.5, Upper: 11.0. 3. Save.	Production reference range is created successfully in LIVE. It is independent of any TEST study ranges. Data entry users in LIVE can now trigger out-of-range flags against this range.
SMG-07	Incorrect deploy sequence (TEST → DEV) is blocked or flagged	<i>A user attempts to push configuration from TEST back to DEV</i>	1. In TEST study, look for an option to deploy back to DEV. 2. Attempt reverse deployment.	System does not allow reverse deployment (TEST → DEV). The deployment flow is one-directional: DEV → TEST → LIVE. Any attempt to reverse is blocked or no such option exists in the deployment UI.
SMG-08	Study Admin in LIVE can still modify lab details post-deployment	<i>LIVE study has 'Production Lab'; address was initially set incorrectly</i>	1. In LIVE, navigate to Lab Management → Production Lab → DETAILS. 2. Update address field. 3. Save.	Lab details (name, code, address) are editable in LIVE after deployment. Only Lab variable configurations (Lab Test Result, Range Determinant settings) are locked post-deployment. Lab operational data remains editable.

Section 18 — Unsupported Features & Negative Scenarios

Validates system behavior when attempting unsupported Lab Management operations such as DDE, MSDE, API, and real-time reports.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
NEG-01	Verify Double Data Entry (DDE) is not supported for Lab forms	<i>Lab form 'CBC Panel' exists; a subject has data entry pending for DDE workflow</i>	1. Attempt to enable or initiate Double Data Entry for the CBC Panel Lab form. 2. Observe system behavior.	DDE is not available for Lab forms. Either the option is grayed out, or the system displays: 'Lab Management does not support Double Data Entry'. Lab form cannot be set up for DDE.
NEG-02	Verify Multiple Subject Data Entry (MSDE) is not supported for Lab forms	<i>MSDE feature is available in the study for non-Lab forms</i>	1. Attempt to use Multiple Subject Data Entry mode for a CBC Panel Lab form. 2. Observe behavior.	MSDE is blocked or unavailable for Lab forms. System displays appropriate restriction message. MSDE can only be used with non-Lab form types.
NEG-03	Verify Real-time reports do not include Lab Management data	<i>Real-time reports feature is enabled; Lab forms exist with data</i>	1. Navigate to real-time reports section. 2. Attempt to include CBC Panel Lab form or Lab-specific columns.	Lab Management data is not available in real-time reports. If Lab forms are selected, Lab-specific flag data, reference ranges, and lab name fields are absent from real-time report output.
NEG-04	Verify BYO (Build Your Own) reports do not support Lab data	<i>BYO reports feature is accessible</i>	1. Navigate to BYO reports builder. 2. Attempt to add Lab-specific fields (Flag State, Reference Range, Lab Name) from Lab forms.	Lab-specific fields are not available in BYO report builder. Standard form fields may be available, but Lab Management-specific data (flags, ranges, lab name) cannot be added to BYO reports.
NEG-05	Import API does not support Lab Management	<i>Import API is configured for the study</i>	1. Attempt to import data via Import API that includes Lab form data with a lab selection. 2. Observe whether Lab field is accepted.	Import API does not support Lab Management. The Lab field is not recognized or processed via Import API. Users must use Bulk Upload or direct data entry for lab data.
NEG-06	Export API does not include Lab Management data	<i>Export API is configured and accessible</i>	1. Call the Export API for a study that has Lab forms with out-of-range flags. 2. Inspect the API response for Lab-specific fields.	Export API does not include Lab Management data fields (Flag State, Reference Range values, Lab Name, Out-of-Range status). These fields are only available through the UI-based Data Export with Include Labs option.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
NEG-07	Lab Test Result variable cannot be set on non-Lab form	A standard (non-Lab) form 'Subject Status' exists	1. Navigate to Configure → Forms → Subject Status. 2. Open any Number variable in Variable Editor. 3. Check if 'Lab Test Result' is available under Lab Data.	'Lab Test Result' setting is NOT available for variables on non-Lab forms. The Lab Data setting only appears in Variable Editor for variables on forms that are designated as Lab form use type.
NEG-08	Verify data entry fields for Lab form cannot be saved without Lab selection (boundary test)	Lab form 'Coagulation Panel' is open for a new subject	1. Enter all required fields (PT = 11, INR = 1.0, aPTT = 30). 2. Leave Lab field empty. 3. Click Save & Next.	System prevents save. Validation message: 'Please select a lab before saving this form.' All entered data is retained in the form pending lab selection.
NEG-09	Entering out-of-range value for an inactive reference range — no flag triggered	Reference range 'WBC Pediatric Range' has been deactivated; a new data entry is being made	1. Open CBC Panel for a new subject. 2. Select lab that had 'WBC Pediatric Range'. 3. Enter WBC = 2.0 (would have triggered the deactivated range). 4. Save.	No out-of-range flag appears for WBC = 2.0 since 'WBC Pediatric Range' is deactivated. Only active reference ranges trigger flags.
NEG-10	Lab Management features are inaccessible when Labs is disabled in EDC Admin	Labs checkbox under EDC Admin → Enable is currently unchecked	1. Navigate to EDC Admin → Enable → uncheck Labs → Save. 2. Navigate to Manage → check for Lab Management menu item. 3. Navigate to Configure → Roles and Permissions → check for Labs section.	Manage menu does not show Lab Management. Lab Management is completely inaccessible. Roles and Permissions does not show Labs section. Disabling EDC Admin Labs flag removes all Lab Management access.

Section 19 — Lab Data Export

Validates Export Labs and Data Export (Include Labs) functionality, output content accuracy, and flag state representation.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LDE2-01	Export Labs produces Excel with lab details and reference range tabs	BioReference Lab and Quest Diagnostics exist with reference ranges configured	1. Navigate to Manage → Lab Management. 2. Click three-dot icon next to Save → 'Export Labs'. 3. Open downloaded Excel file.	Excel file downloads. Contains a 'Lab Details' tab listing all labs (name, code, address). Each lab has its own reference range tab with columns for variable name, range name, lower/upper limits, determinants, units, LOINC.
LDE2-02	Data Export with Include Labs — all 8 required fields present	CBC Panel Lab form has data for 10 subjects; out-of-range flags triggered for some	1. Navigate to Analysis → Data Export. 2. Select 'CBC Panel' Lab form. 3. Check 'Include Labs'. 4. Click Download data export file. 5. Open Excel and inspect columns.	Export columns include: Lab Name, Flag State, Reference Range values, Range Determinant values, Out-of-Range, Lower Limit, Upper Limit, Expected Value. All 8 required fields are present.
LDE2-03	Flag State values are correct in Data Export	Subjects have varying WBC values: within range, above range, below range	1. Export data with Include Labs. 2. Inspect the 'Out-of-Range' column.	Out-of-Range column contains exactly: 'Within', 'Above', or 'Below' for numeric variables, and 'Equal' or 'Not Equal' for categorical variables. No other values appear.
LDE2-04	Export Labs includes deactivated reference ranges	Reference range 'Hemoglobin Pediatric Range' was deactivated	1. Export Labs → open Excel → navigate to BioReference Lab tab. 2. Check for deactivated range in the list.	Deactivated reference range 'Hemoglobin Pediatric Range' appears in the export with a status indicator (e.g., Inactive). Historical ranges are included for compliance tracking.
LDE2-05	System does not warn about conflicting ranges in UI — only discoverable via export	WBC Count has overlapping ranges: Range A 3.0–7.0 and Range B 5.0–10.0	1. Set up overlapping ranges. 2. Navigate through Lab Management UI → observe for any conflict warnings. 3. Export Labs → review range definitions.	UI does NOT display any conflict or overlap warning. Export file contains both ranges listed. QA must manually review export to identify overlaps. This is documented expected behavior.

TC-ID	Test Case Title	Pre-Conditions	Test Steps	Expected Result
LDE2-06	Data Export without Include Labs does not contain lab-specific columns	CBC Panel Lab form has data and lab-specific columns	1. Navigate to Analysis → Data Export. 2. Select CBC Panel. 3. Do NOT check 'Include Labs'. 4. Download and inspect.	Export does not include: Lab Name, Flag State, Out-of-Range, Reference Range values. Only standard clinical data columns are present. Include Labs must be explicitly checked for lab data.
LDE2-07	Export includes LOINC codes and units in Lab exports	WBC Count has LOINC '26464-8' and unit 'K/uL' configured	1. Export Labs → open Excel → navigate to BioReference Lab tab. 2. Locate WBC Count reference range row. 3. Check LOINC and Unit columns.	WBC Count row in export shows Unit = 'K/uL' and LOINC = '26464-8' in the corresponding columns.
LDE2-08	Data Export flag state is 'Equal' for categorical Yes/No exact match	X-Ray Taken: Expected = Yes; Subject S-040 entered Yes	1. Export data with Include Labs for Imaging Panel. 2. Find X-Ray Taken row for S-040. 3. Check Out-of-Range column.	Out-of-Range column shows 'Equal' for subject S-040 who entered 'Yes' (matching the expected value). No flag triggered and export correctly reflects 'Equal' status.